

Human Reproduction

Year 9 Science · exam sprint · 2 hours

- 0:04 Loop 1 · Sex cells and the two systems
- 0:26 Loop 2 · Hormones and the menstrual cycle
- 0:49 Loop 3 · Fertilisation, chromosomes, twins
- 1:12 Loop 4 · Pregnancy, birth, fertility
- 1:44 Mini mock

Human Reproduction 人类生殖

Year 9 科学 · 考前冲刺 · 2 小时，中间不休息

0:00	开场：Do Now 和核心思想	第 1–3 页
0:04	Loop 1 · 性细胞和男女生殖系统	第 4–14 页
0:26	Loop 2 · 激素和月经周期	第 15–24 页
0:49	Loop 3 · 受精、染色体、双胞胎	第 25–35 页
1:12	Loop 4 · 怀孕、出生、不孕	第 36–49 页
1:42	失分点和总结	第 50–51 页
1:44	Mini mock (In-class work 的 Part E)	第 52 页
1:59	Homework	第 53 页

老师页

Dr. Ryan Lei · Auckland 1

开课前发 In-class work (Part A–E，一份纸)；Homework 下课时发。每个 loop 讲完马上做对应的 Part，约 6 分钟。页面上的红字是答案，右边蓝字是读音，大写的音节读重音。

- Say it 读音
- reproduction
ree-pruh-DUK-shun
- hormones
HOR-mohnz
- menstrual
MEN-stroo-ul
- fertilisation
fer-tih-ly-ZAY-shun
- chromosomes
KROH-muh-sohmz
- pregnancy
PREG-nun-see
- fertility
fer-TIL-ih-tee

Do Now

- 1 Name the male sex cell and the female sex cell.
- 2 Which organ makes the female sex cells?
- 3 What is the joining of the two sex cells called?
- 4 How many chromosomes are in a human body cell?
- 5 Which organ passes oxygen from the mother to the foetus?

Do Now 课前小测

- 1 说出男性和女性的 sex cell (性细胞)。
male: sperm (精子); female: egg / ovum (卵)
- 2 哪个器官产生女性的性细胞？
the ovaries (卵巢)
- 3 两个性细胞结合，叫什么？
fertilisation (受精)
- 4 人的一个体细胞里有多少条 chromosomes (染色体)？
46 (23 pairs, 23 对)
- 5 哪个器官把氧气从母亲传给 foetus (胎儿)？
the placenta (胎盘)

0:00–0:02。让她口答，先不讲解。第 1–3 题错：Loop 1 放慢；第 4 题错：Loop 3 的染色体页多停；第 5 题错：Loop 4 的 placenta 页多停。

Say it 读音

ovum

OH-vum

ovaries

OH-vuh-reez

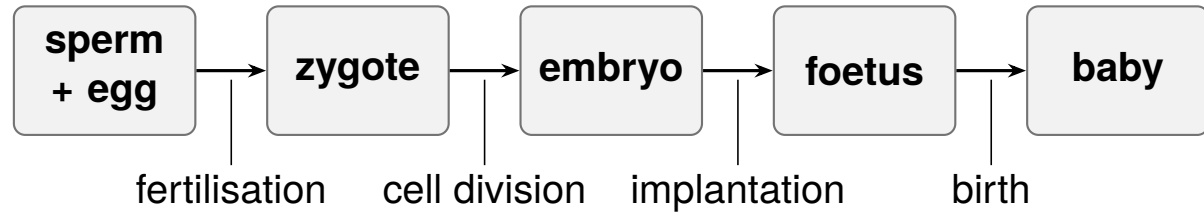
placenta

pluh-SEN-tuh

foetus

FEE-tus

The one idea: from two cells to a baby

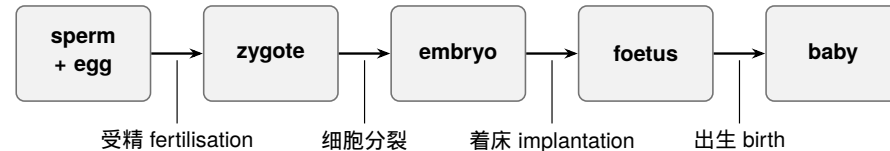


Every question is about one of these steps.

For each step, know the cell, the place and the process.

Which step happens in the oviduct?

The one idea 核心思想：从两个细胞到一个婴儿



每道题都在问这条线上的某一步。

每一步要知道三样：是哪个细胞、在哪里、叫什么过程。

哪一步发生在 oviduct (输卵管) 里？

fertilisation happens in the oviduct; zygote 一边分裂一边向 uterus (子宫) 移动

0:02–0:04。指着图从左到右把英文词念一遍。强调：fertilisation 在 oviduct; implantation (着床) 才在 uterus。这是全单元最常见的失分点。

Say it 读音

zygote

ZY-goht

embryo

EM-bree-oh

implantation

im-plan-TAY-shun

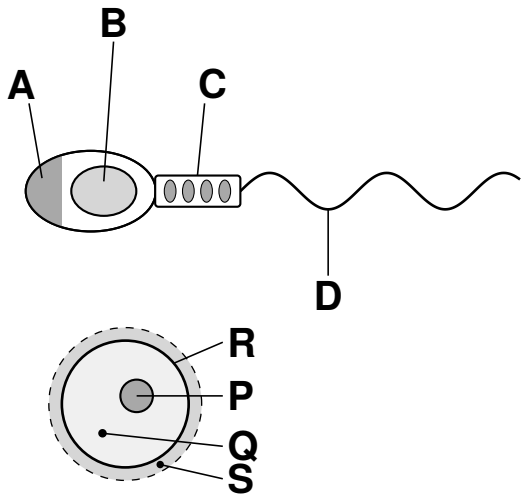
oviduct

OH-vih-dukt

uterus

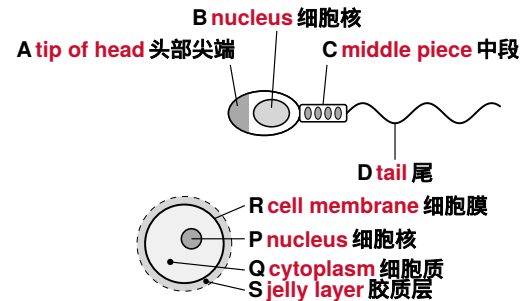
YOO-tuh-rus

Quick review: Sex cells



- 1 Name parts A to D and parts P to S.
- 2 How many chromosomes are in each nucleus?
- 3 Which cell is larger? Which cell can move?

Quick review 快速回忆：性细胞



- 1 说出 A-D 和 P-S 的名称。
见图上红字
- 2 每个 nucleus 里有多少条 chromosomes?
23 in each
- 3 哪个细胞更大？哪个会动？
the egg is much larger;
only the sperm can move:
it swims with its tail

0:04-0:06。A (头部尖端) 里有 enzymes (酶) ; C (中段) 里有 mitochondria (线粒体 , 供能) 。
图不按比例 : egg 实际比 sperm 大得多。

Say it 读音

nucleus

NYOO-klee-us

cytoplasm

SY-toh-plaz-um

membrane

MEM-brayn

enzymes

EN-zymz

mitochondria

my-toh-KON-dree-uh

Adaptations: the feature and how it helps

Cell	Feature	How it helps
sperm	long tail	
sperm	many mitochondria	
sperm	enzymes in the head	
egg	large cytoplasm	
egg	membrane changes	

Why does a man make millions of sperm, but a woman releases one egg a month?

Adaptations 适应：特征 + 有什么用

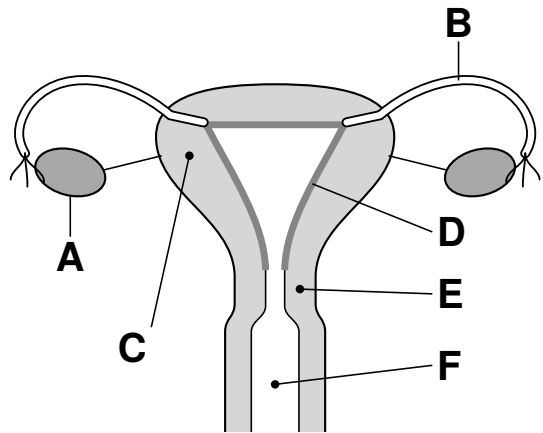
细胞	Feature 特征	How it helps 有什么用
sperm	long tail 长尾	<u>it swims to the egg</u>
sperm	很多 mitochondria	<u>they release energy for swimming</u>
sperm	头部有 enzymes	<u>they break through the jelly layer and membrane</u>
egg	cytoplasm 很大	<u>a food store for the first cell divisions</u>
egg	membrane 会变化	<u>it stops any other sperm getting in</u>

为什么男性产生几百万个 sperm，而女性每月只排一个 egg？
very few sperm survive the journey to the oviduct, so millions give a better chance that one meets the egg

0:06–0:08。Adaptation 题的写法：feature + how it helps。她 booklet p.5 的表只写了特征 (large、swims with tail)，没写 “so that ...”，这样只能拿 A (L3)。

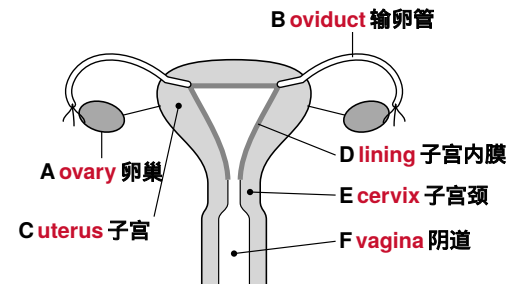
Say it 读音
adaptations
ad-ap-TAY-shunz

Quick review: Female system, front view



- 1 Name parts A to F.
- 2 Where are eggs made?
- 3 Where does a foetus develop?
- 4 Which part is a ring of muscle?

Quick review 快速回忆：女性生殖系统，正面图



- 1 说出 A-F 的名称。
见图上红字。B = fallopian tube; D = endometrium
- 2 卵在哪里产生？
in the ovaries
- 3 胎儿在哪里发育？
in the uterus
- 4 哪一部分是一圈肌肉？
the cervix

0:08-0:09。她在 booklet p.8 的正面图上把 uterus 和 cervix 标反了：cervix 是 uterus 底部最窄的那一段。让她指着屏幕再说一遍。

Say it 读音

ovary
OH-vuh-ree

fallopian
fuh-LOH-pee-un

endometrium
en-doh-MEE-tree-um

cervix
SUR-viks

vagina
vuh-JY-nuh

Female system: the job of each part

Part	Job
ovary	makes _____ and two hormones
oviduct	carries the egg; _____ happens here
uterus	the _____ develops here
lining	gets thick so that an _____ can implant
cervix	a ring of _____ at the base of the uterus
vagina	receives _____; the baby is born through it

女性生殖系统：每部分的作用

Part 部位	Job 作用
ovary 卵巢	产生 <u>eggs</u> 和两种 hormones (oestrogen、 progesterone)
oviduct 输卵管	运送 egg; <u>fertilisation</u> 在这里发生
uterus 子宫	<u>foetus</u> 在这里发育
lining 子宫内膜	变厚, 让 <u>embryo</u> 可以 implant (着床)
cervix 子宫颈	uterus 底部的一圈 <u>muscle</u>
vagina 阴道	接收 <u>sperm</u> (semen); 婴儿从这里出生

0:09–0:10。让她一行一行说出空格里的词。“fertilisation 在 oviduct”让她说两遍。

Say it 读音

oestrogen

EE-struh-jun

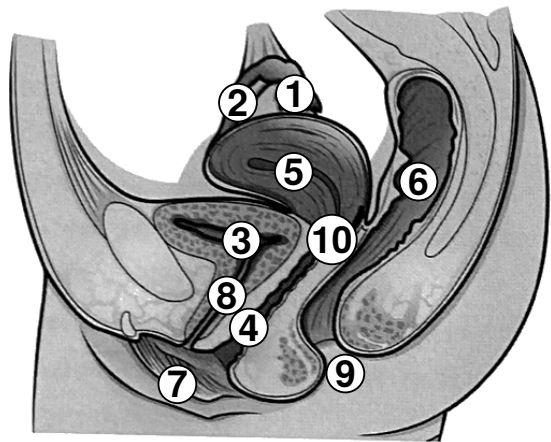
progesterone

proh-JES-tuh-rohn

semen

SEE-mun

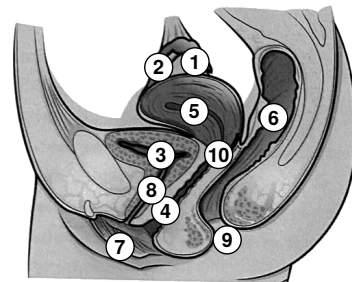
Quick review: Female system, side view



anus · bladder · cervix ·
fallopian tube · ovary · rectum ·
urethra · uterus · vagina · vulva

- 1 Match the words to parts 1 to 10.
- 2 Which four parts are not reproductive organs?

Quick review 快速回忆：女性生殖系统，侧面图



anus · bladder · cervix · fallopian tube · ovary ·
rectum · urethra · uterus · vagina · vulva

- 1 把这些词和 1–10 对上。
1 ovary 卵巢 2 fallopian tube 输卵管 3 bladder 膀胱 4 vagina 阴道 5 uterus 子宫
6 rectum 直肠 7 vulva 外阴 8 urethra 尿道 9 anus 肛门 10 cervix 子宫颈
- 2 哪四个不是生殖器官？
bladder, urethra (urine 尿); rectum, anus (消化系统)

0:10–0:12. booklet p.8 这张图没有印答案，以上是按解剖位置判读的：3 在最前面，是 bladder；6 在最后面，是 rectum；10 在 uterus 和 vagina 之间，是 cervix。

Say it 读音

anus

AY-nus

rectum

REK-tum

urethra

yoo-REE-thruh

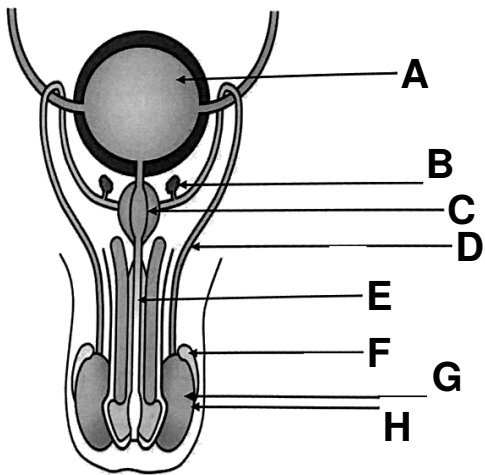
vulva

VUL-vuh

urine

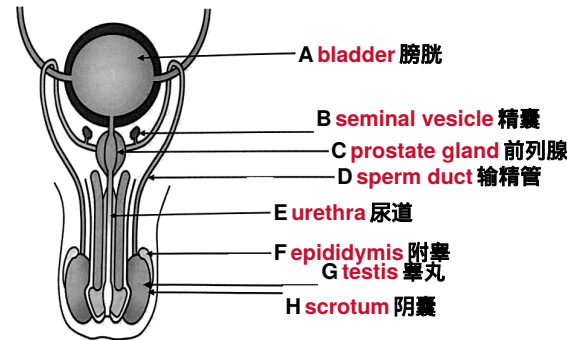
YOOR-in

Quick review: Male system



- 1 Name parts A to H.
- 2 Where are sperm made?
- 3 Which tube carries both urine and semen?

Quick review 快速回忆：男性生殖系统



- 1 说出 A-H 的名称。
见图上红字。D = vas deferens
- 2 sperm 在哪里产生？
in the testes
- 3 哪条管道既走 urine 又走 semen？
the urethra

0:12–0:13。G、H 两个箭头挨得很近：上面指 testis，下面指 scrotum (booklet 没印答案，按位置判读)。A bladder 不是生殖器官。

Say it 读音

seminal
SEM-ih-nul

vesicle
VES-ih-kul

prostate
PROS-tayt

vas deferens
vas-DEF-uh-renz

epididymis
ep-ih-DID-ih-mis

testis
TES-tis

testes
TES-teez

scrotum
SKROH-tum

The path of a sperm cell

Put the cards in order.

- A** The urethra carries semen out through the penis.
- B** Sperm are made in a testis.
- C** Sperm travel along the sperm duct.
- D** Sperm are stored in the epididymis.
- E** Glands add fluid: sperm + fluid = _____.

Why are the testes held outside the body?

一个 sperm cell 走的路

把卡片排好顺序。

- A** urethra 把 semen 经 penis 送出体外。
- B** sperm 在 testis 里产生。
- C** sperm 沿着 sperm duct 走。
- D** sperm 储存在 epididymis 里。
- E** glands (腺体) 加入液体: sperm + fluid = **semen**。

顺序: B → D → C → E → A

为什么 testes 要放在体外?

sperm are made best a little below body temperature; in the scrotum the testes stay cooler

0:13–0:15. 先让她排序, 再问 sperm 和 semen 的区别: semen = sperm + 腺体加的液体。

Say it 读音

penis

PEE-nis

Male system: the job of each part

Part	Job
scrotum	holds the testes outside the body, so they stay _____
testes	make _____ and the hormone _____
epididymis	_____ the sperm
sperm duct	carries sperm to the _____
glands	add _____ that feeds the sperm
penis	puts semen into the _____

男性生殖系统：每部分的作用

Part 部位	Job 作用
scrotum 阴囊	把 testes 放在体外，让它们保持 cooler
testes 睾丸	产生 sperm 和激素 testosterone
epididymis 附睾	stores sperm
sperm duct 输精管	把 sperm 运到 urethra
glands 腺体	加入 fluid ，给 sperm 提供养分
penis 阴茎	把 semen 送进 vagina

0:15–0:16。glands 指 seminal vesicles 和 prostate gland。她 booklet p.11 的这张表写得不错，这一页快速过。

Say it 读音

testosterone

tes-TOS-tuh-rohn

Spot the error (1)

Describe where sex cells are made and where a foetus develops.

Sam's answer

Eggs are made in the uterus and travel down the oviduct. The cervix is the place where the foetus develops. In a man, sperm are made in the testes and stored in the scrotum.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error 找错 (1)

题目：说出性细胞在哪里产生，胎儿在哪里发育。

Sam's answer

Eggs are made in the uterus and travel down the oviduct. The cervix is the place where the foetus develops. In a man, sperm are made in the testes and stored in the scrotum.

Sam 的答案有 3 处错。找出来，改正，并说出为什么丢分。

1. eggs are made in the ovaries, not in the uterus (L4)
2. the foetus develops in the uterus; the cervix is the ring of muscle at its base (L4)
3. sperm are stored in the epididymis; the scrotum is the bag of skin that holds the testes (L4)

0:16–0:17。“travel down the oviduct” 和 “sperm are made in the testes” 是对的。让她一句一句判断，不要看到 Sam 就整段改。

Practice

Q1 State two differences between a sperm cell and an egg cell. **[A]**

Q2 Explain why the testes are held outside the body in the scrotum. **[M]**

Q3 A woman's oviducts are both blocked. She still releases an egg each month. Explain why she cannot become pregnant naturally. **[E]**

Practice 练习

Q1 说出 sperm cell 和 egg cell 的两个不同点。 **[A]**

any two: a sperm is much smaller; a sperm has a tail and swims, an egg cannot move; an egg has a large food store; millions of sperm, one egg a month

Q2 解释为什么 testes 在体外的 scrotum 里。 **[M]**

sperm are made best a little below body temperature; outside the body the testes stay cooler

Q3 一位女性两侧的 oviducts 都堵了，但每月仍然排卵。解释她为什么不能自然怀孕。 **[E]**

fertilisation happens in the oviduct; the sperm cannot reach the egg and the egg cannot reach the uterus, so no zygote forms and nothing can implant

0:17–0:20. 让她先口答，再对红字。Q3 要说到“fertilisation 在 oviduct”才算 E。

Classwork

In-class work · Part A

- 1 Label the female reproductive system.
- 2 Complete the sentences about the male system.
- 3 Explain one adaptation of a sperm cell.

Write in full sentences. Use the science words.

Classwork 课堂练习

In-class work · Part A (约 6 分钟)

- 1 给女性生殖系统的图标注。
- 2 补全关于男性生殖系统的句子。
- 3 解释 sperm cell 的一个 adaptation。

要求：写完整的句子，用科学词。

答案在 **inclass_answers.pdf** 的 **Part A**

Puberty: hormones switch it on

Hormones are chemical _____, carried in the _____.

Boys	Girls
hormone _____	_____
made in testes	ovaries
changes voice deepens	_____ start
also sperm are made	breasts grow

True or false? Girls usually start puberty before boys.

Puberty 青春期：由 hormones 启动

Hormones (激素) 是化学 messengers，随 blood 运到全身。

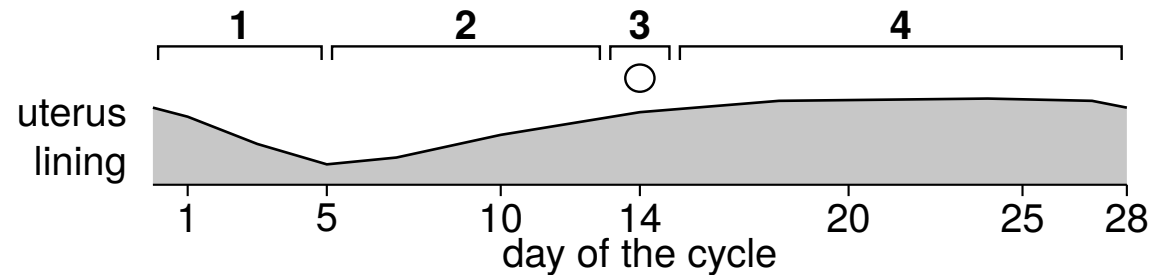
	男孩	女孩
hormone 激素	<u>testosterone</u>	<u>oestrogen</u>
产生于	testes	ovaries
变化	声音变低	<u>periods</u> 开始
另外	开始产生 sperm	乳房发育

判断对错：女孩通常比男孩先进入 puberty。 **True**

0:26–0:28。男女都有的变化：长高，长阴毛和腋毛。booklet p.6 的 17 道判断题她做过了，这里不展开。

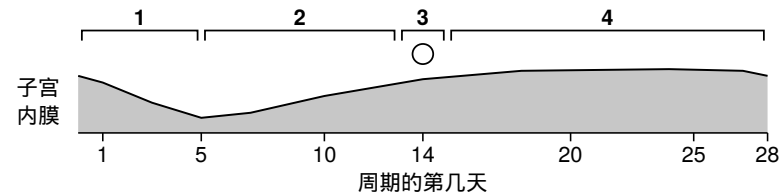
Say it 读音
puberty
PYOO-buh-tee

The menstrual cycle: stages 1 and 2



- 1 Days 1 to 5: the lining _____ and leaves through the vagina. This is a _____.
- 2 Days 5 to 13: the lining _____ again. An egg matures in an _____.

Menstrual cycle 月经周期：第 1、2 阶段



- 1 第 1-5 天：内膜 **breaks down**，经 vagina 流出。这就是 **period (menstruation)**。
 - 2 第 5-13 天：内膜重新 **builds up**；一个 egg 在 **ovary** 里成熟。
- 阶段名称：1 **menstrual** 2 **follicular**

0:28-0:30。图上方的 1-4 是四个阶段，灰色表示内膜的厚度，小圆圈是第 14 天排出的 egg。先让她说每一段内膜是变厚还是变薄。第 2 阶段 ovary 里的 follicle (卵泡) 长大，FSH 让里面的 egg 成熟。

Say it 读音

menstruation

men-stroo-AY-shun

follicular

fuh-LIK-yoo-luh

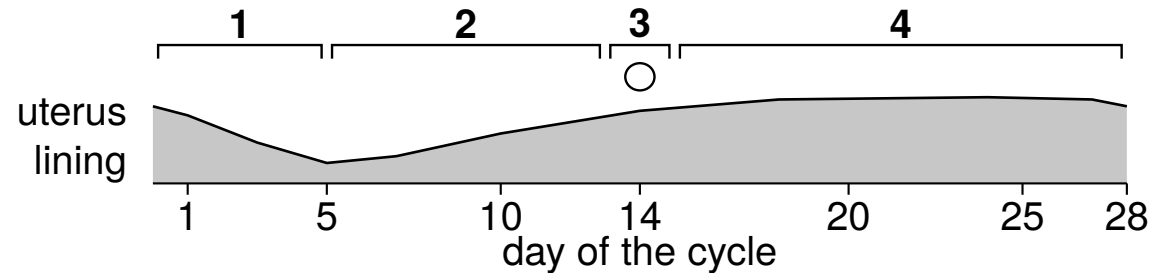
follicle

FOL-ih-kul

stimulating

STIM-yoo-lay-ting

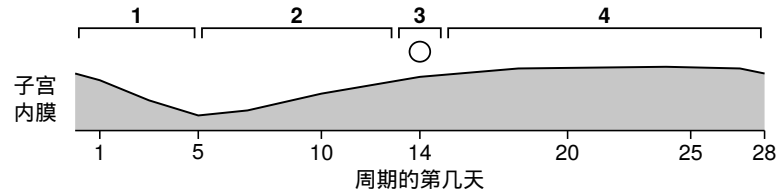
The menstrual cycle: stages 3 and 4



3 About day 14: an egg is _____ from an ovary. This is _____.

4 Days 15 to 28: the lining stays _____, ready for a fertilised egg.
When is a woman most likely to become pregnant?

Menstrual cycle 月经周期：第 3、4 阶段



3 大约第 14 天：一个 egg 从 ovary **released**。这叫 **ovulation** (排卵)。

4 第 15–28 天：内膜保持 **thick**，等待受精卵。

阶段名称：3 ovulation 4 luteal。没有受精卵：内膜 breaks down，周期重新开始

她什么时候最容易怀孕？

around day 14: the day of ovulation and the two days before and after it

0:30–0:32。易错 (L5)：ovulation 是排卵，约第 14 天；menstruation 是内膜脱落，从第 1 天开始。两个词不要混。

Say it 读音

ovulation

ov-yoo-LAY-shun

luteal

LOO-tee-ul

Four hormones, four jobs

Hormone	Made in	Job
FSH	pituitary gland	makes an egg _____ in the ovary
LH	_____	triggers _____ on day 14
oestrogen	ovary	_____ the lining of the uterus
progesterone	_____	keeps the lining _____

FSH = follicle stimulating hormone. LH = luteinising hormone.
Which two hormones peak at ovulation?

四种 hormones, 四个作用

Hormone	产生于	作用
FSH	pituitary gland (垂体)	让一个 egg 在 ovary 里 mature
LH	pituitary gland	在第 14 天触发 ovulation
oestrogen	ovary	builds up 子宫内膜
progesterone	ovary	让内膜保持 thick

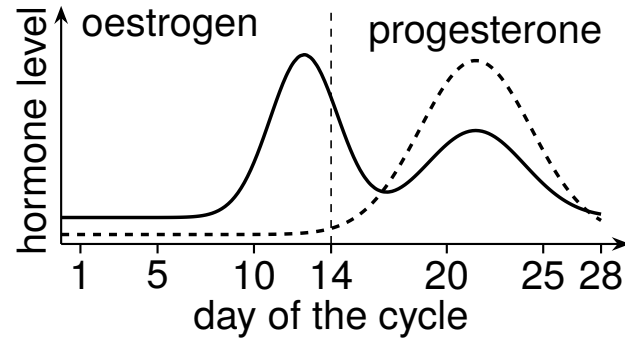
FSH = follicle stimulating hormone. LH = luteinising hormone.
哪两种激素在 ovulation 时达到峰值？ **FSH and LH**

Booklet p.13–14 Dr. Ryan Lei · Auckland 18

0:32–0:34。记法：垂体的两个 (FSH、LH) 管 egg；卵巢的两个 (oestrogen、progesterone) 管内膜 (L6)。她 booklet p.16 第 5 题写了两个激素，标准说法是 progesterone 下降，内膜脱落。

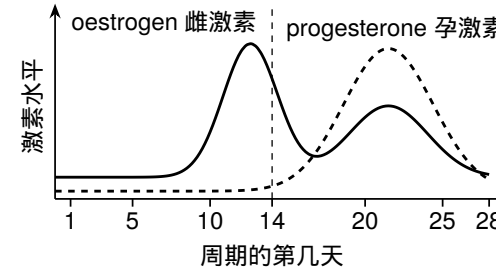
Say it 读音
pituitary
pih-TYOO-ih-tree
luteinising
LOO-tee-in-eye-zing

Read the graph: ovarian hormones



- 1 Which hormone peaks just before day 14?
- 2 Which hormone is highest at about day 21?
- 3 Both levels fall by day 28. What happens to the lining?

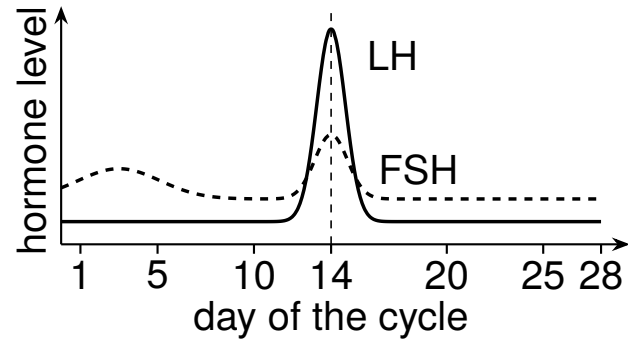
读图：ovaries 产生的激素



- 1 哪种激素在第 14 天之前达到峰值？
oestrogen
- 2 第 21 天左右哪种激素最高？
progesterone
- 3 到第 28 天两种都下降了。内膜会怎样？
the lining breaks down and is lost: a period starts

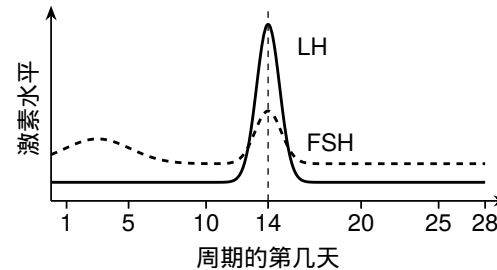
0:34–0:36. 读图题让她先说“哪条线、第几天、高还是低”，再说原因。实线是 oestrogen，虚线是 progesterone。

Read the graph: pituitary hormones



- 1 On which day do LH and FSH peak?
- 2 What does the LH peak cause?
- 3 Describe the level of LH from day 1 to day 12.

读图：pituitary gland 产生的激素



- 1 LH 和 FSH 在第几天达到峰值？
day 14
- 2 LH 的峰值引起什么？
ovulation: an egg is released from an ovary
- 3 描述第 1–12 天 LH 的水平。
it stays low and almost constant

0:36–0:37. 第 3 题的指令词是 describe: 只说趋势, 不要解释原因 (L12)。实线是 LH, 虚线是 FSH。

No fertilised egg: write it in this order

- 1 The egg is not fertilised, so no embryo _____ in the lining.
- 2 The level of _____ falls.
- 3 The lining of the uterus _____.
- 4 Blood and lining leave through the _____: a period.

A: lines 3–4 **M:** links the hormone to the lining **E:** the whole chain

What is different if the egg is fertilised?

卵没有受精：按这个顺序写

- 1 卵没有受精，所以没有 embryo **implants** 到内膜里。
- 2 **progesterone** 的水平下降。
- 3 子宫内膜 **breaks down**。
- 4 血和内膜经 **vagina** 排出：这就是 period。

A: 第 3–4 行 **M:** 把激素和内膜连起来 **E:** 整条链

如果卵受精了，有什么不同？

the embryo implants, progesterone stays high and the lining stays thick, so there is no period

0:37–0:39. Explain 题按 1 到 4 的顺序写。只写 “a period starts” 是 A；写出 progesterone falls, lining breaks down 才到 M。

Spot the error (2)

Describe ovulation and name one hormone that changes the lining of the uterus.

Sam's answer

Ovulation is when the lining of the uterus is lost. An egg is released on about day 28 of the cycle. Oestrogen is made in the pituitary gland and makes the lining thicker.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error 找错 (2)

题目：描述 ovulation，并说出一种改变子宫内膜的激素。

Sam's answer

Ovulation is when the lining of the uterus is lost. An egg is released on about day 28 of the cycle. Oestrogen is made in the pituitary gland and makes the lining thicker.

Sam 的答案有 3 处错。 找出来，改正，并说出为什么丢分。

- 1. ovulation is the release of an egg from an ovary; losing the lining is menstruation (L5)**
- 2. the egg is released on about day 14, not day 28 (L5)**
- 3. oestrogen is made in the ovaries, not in the pituitary gland (L6)**

0:39–0:40。最后半句“makes the lining thicker”是对的。

Practice

Q1 A woman has a 28-day cycle. On which days is she most likely to become pregnant? **[A]**

Q2 Describe how the lining of the uterus changes between day 5 and day 14. Name the hormone that causes this. **[M]**

Q3 The level of progesterone stays high in a pregnant woman. Explain why this is important. **[E]**

Practice 练习

Q1 一位女性的周期是 28 天。她在哪几天最容易怀孕？ **[A]**
about days 12 to 16: ovulation is on about day 14, and she is fertile for two days before and after it

Q2 描述第 5 天到第 14 天子宫内膜的变化，并说出引起变化的激素。 **[M]**
the lining builds up and becomes thicker, with more blood vessels; oestrogen

Q3 孕妇的 progesterone 水平一直很高。解释为什么这很重要。 **[E]**
progesterone keeps the lining thick, so it is not lost; the embryo stays implanted and the placenta can form, so there are no periods during pregnancy

0:40–0:43. Q3 是 E：要写到 “so the embryo stays implanted”。

Classwork

In-class work · Part B

- 1 Name the four stages of the cycle.
- 2 Match each hormone to its job.
- 3 Read a table of hormone levels and explain it.

For data: describe the trend first, with numbers.

Classwork 课堂练习

In-class work · Part B (约 6 分钟)

- 1 说出周期的四个阶段。
- 2 把每种激素和它的作用连起来。
- 3 读一张激素水平的数据表并解释。

数据题：先描述趋势，带上数字。

答案在 [inclass_answers.pdf](#) 的 Part B

From intercourse to fertilisation

Put the cards in order.

- A** Sperm swim through the cervix and uterus into the oviducts.
- B** The penis fills with blood and becomes erect.
- C** One sperm enters the egg, and the two nuclei fuse.
- D** Semen is released into the vagina. This is ejaculation.
- E** The penis is put into the vagina.

Which card is fertilisation? Where does it happen?

从 intercourse 到 fertilisation

把卡片排好顺序。

- A** sperm 游过 cervix 和 uterus, 进入 oviducts.
- B** penis 充血, 变硬 (erect)。
- C** 一个 sperm 进入 egg, 两个 nuclei (细胞核) 融合。
- D** semen 被射入 vagina。这叫 ejaculation (射精)。
- E** penis 放入 vagina。

顺序: B → E → D → A → C

哪张卡片是 fertilisation? 它在哪里发生?

card C; it happens in the oviduct

0:49–0:51。这是学习目标第 12 条。当成一道普通的排序题来做, 按生物过程平铺直叙就行。nuclei 是 nucleus 的复数, 读 NYOO-klee-eye。

Say it 读音

intercourse

IN-ter-kors

ejaculation

ih-jak-yoo-LAY-shun

Zygote, embryo, foetus

- 1 The sperm nucleus fuses with the egg nucleus. The new cell is a _____.
- 2 It divides many times as it moves along the oviduct. The ball of cells is an _____.
- 3 The ball of cells sinks into the lining of the uterus. This is _____.
- 4 After about 8 weeks it has all its organs. Now it is a _____.

What stops a second sperm from entering the egg?

Zygote、embryo、foetus

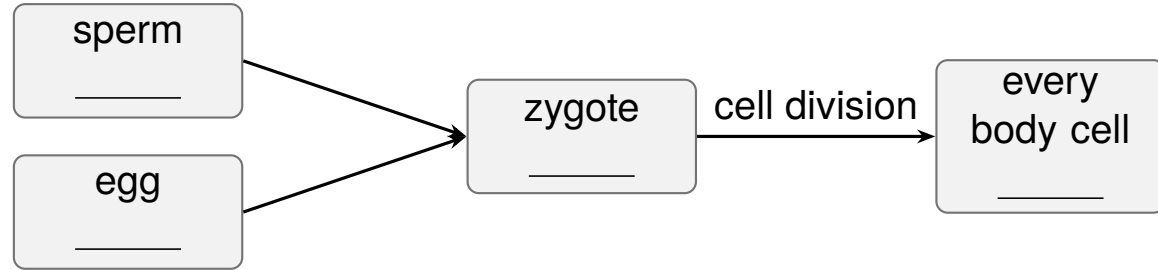
- 1 sperm 的 nucleus 和 egg 的 nucleus 融合。这个新细胞叫 zygote (受精卵)。
- 2 它沿着 oviduct 移动，同时不断分裂。这团细胞叫 embryo (胚胎)。
- 3 这团细胞陷进子宫内膜。这叫 implantation (着床)。
- 4 大约 8 周后，器官都有了。这时叫 foetus (胎儿)。

什么阻止第二个 sperm 进入 egg ?

the membrane of the egg changes as soon as one sperm has entered

0:51–0:53. 易错 (L8): 顺序是 zygote, embryo, foetus; implantation 不是 fertilisation。她 booklet p.17 的 fertilisation 定义还空着，让她写：the nucleus of a sperm fuses with the nucleus of an egg。

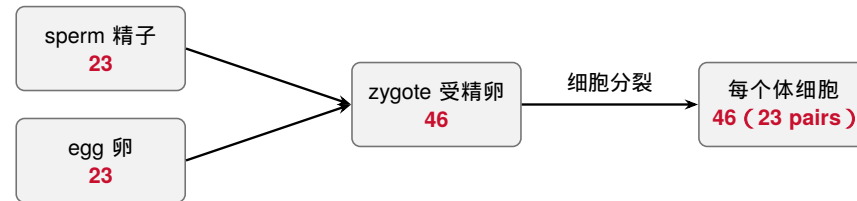
Chromosomes: count them



Chromosomes carry the genes. They are in the _____ of every cell.

Why must a sex cell have only half the number?

Chromosomes 染色体：数一数



Chromosomes 带着 genes (基因), 在每个细胞的 nucleus 里。

为什么 sex cell 只能有一半的数目？

so that the zygote has the full 46 and not 92 when the two sex cells join; half comes from each parent

0:53–0:55。易错 (L1): sperm 和 egg 各 23 条, 不是 46 条。

Boy or girl?

father's sperm

mother's eggs	
X	X
father's sperm	X
	Y

Female cells have XX. Male cells have XY.
Every egg carries _____. A sperm carries _____ or _____.

- 1 Complete the square.
- 2 What is the chance of a boy?
- 3 Which parent's sex cell decides?

生男还是生女？

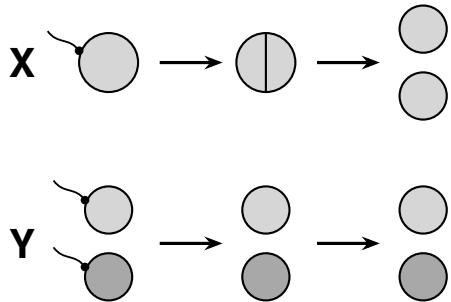
母亲的 eggs	
X	X
父亲的 sperm	X
	Y

女性的细胞是 XX，男性的细胞是 XY。
每个 egg 都带 X。sperm 带 X 或 Y。

- 1 把方格填完。
见左图红字
- 2 生男孩的概率是多少？
50 % (2 of the 4 boxes)
- 3 哪一方的性细胞决定性别？
the father's: a sperm with X gives a girl, a sperm with Y gives a boy

0:55–0:57. 易错 (L7): 决定性别的是 sperm，不是 egg。

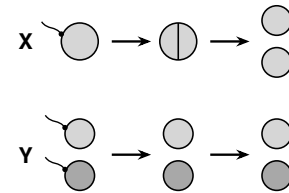
Twins



Each row ends with two embryos.

- 1 Which row shows identical twins?
- 2 Row X: _____ egg, _____ sperm.
The zygote _____.
- 3 Row Y: _____ eggs, _____ sperm.
- 4 Which twins can be a boy and a girl?

Twins 双胞胎



每一行最后都是两个 embryos。

- 1 哪一行是 identical twins (同卵双胞胎)?

row X

- 2 X 行: **one** 个 egg, **one** 个 sperm. zygote **splits into two**。

- 3 Y 行: **two** 个 eggs, **two** 个 sperm。

- 4 哪种双胞胎可以是一男一女?

row Y, the non-identical (fraternal) twins: they have different genes. Identical twins have the same genes, so they are always the same sex.

Booklet p.19

Dr. Ryan Lei · Auckland 29

0:57–0:59. 图里同一种灰色表示基因相同。Y 行的两个 egg 颜色不同, 表示基因不同 (L11)。

Say it 读音

identical

eye-DEN-tih-kul

fraternal

fruh-TUR-nul

Workbook: true or false?

- a** Identical twins happen when a fertilised egg cell divides in two, but the two cells split apart.
- b** Identical twins will always be different sexes.
- c** Non-identical twins happen when two egg cells are fertilised.
- d** Non-identical twins can be the same sex or different sexes.
- e** Identical twins happen when two egg cells are fertilised.
- f** Non-identical twins happen when egg cells turn into embryos on their own.

Workbook 练习册：对还是错？

- a** 同卵双胞胎：一个受精卵分裂成两个，而且两个细胞分开了。 **True**
- b** 同卵双胞胎一定是不同性别。 **False: always the same sex**
- c** 异卵双胞胎：两个卵细胞都受精了。 **True**
- d** 异卵双胞胎可以是同性别，也可以是不同性别。 **True**
- e** 同卵双胞胎：两个卵细胞受精。 **False: that gives non-identical twins**
- f** 异卵双胞胎：卵细胞自己变成胚胎。 **False: each egg must be fertilised by a sperm**

0:59–1:00。这是 booklet p.19 她空着的原题，屏幕上英文原文。让她直接把 true / false 写到 booklet 上。
上面两张图：一个 zygote 的那组是 identical，两个 zygote 的那组是 fraternal。

Workbook: fertilisation

Q1 Where does fertilisation occur?

Q2 Only one sperm cell joins with the egg cell. What do you think happens to all the other sperms?

Q3 Humans have internal fertilisation as a result of moving onto the land in our evolutionary history. Why do you think this is an advantage on the land?

Workbook 练习册: fertilisation

Q1 fertilisation 在哪里发生？

in the oviduct (fallopian tube)

Q2 只有一个 sperm cell 和 egg cell 结合。你认为其他的 sperm 怎样了？

they die and are broken down

Q3 人类在进化中上了陆地，变成 internal fertilisation (体内受精)。为什么这在陆地上是优势？

on land the sex cells would dry out; inside the body the sperm can swim in fluid, the sex cells are protected, and a sperm is much more likely to meet the egg

1:00–1:02。booklet p.18 她空着的三道题，屏幕上英文原文，让她把答案写到 booklet 上。Q3 是 E：要说出“不会干掉”和“更容易相遇”两点。

Say it 读音
evolutionary
ee-vuh-LOO-shun-ree

A boy next time? Write it in this order

A couple have three girls. What is the chance that their fourth child is a boy?

- 1 Every egg carries an _____ chromosome.
- 2 Half of the sperm carry X and half carry _____.
- 3 So at every fertilisation the chance of a boy is _____.
- 4 The children born before do not change the _____.

A: the number **M:** lines 1–2 **E:** line 4 as well

下一个是男孩吗？按这个顺序写

一对夫妇有三个女儿。第四个孩子是男孩的概率是多少？

- 1 每个 egg 都带一条 X 染色体。
- 2 一半的 sperm 带 X，一半带 Y。
- 3 所以每一次受精，生男孩的概率都是 50 %。
- 4 以前生的孩子不会改变这个 chance。

A: 说出数字 **M:** 第 1–2 行 **E:** 再加上第 4 行

1:02–1:03. booklet p.20 的原话：头两个都是男孩，下一个是男孩的概率仍然是 50 %。每次受精都是一个新的 sperm。

Spot the error (3)

Explain where a baby's chromosomes come from, and how twins can be identical.

Sam's answer

A sperm has 46 chromosomes and an egg has 46, so the baby gets all of them. The mother's egg decides if the baby is a boy or a girl. Identical twins form when two eggs are fertilised by two sperm.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error 找错 (3)

题目：解释婴儿的染色体从哪里来，以及双胞胎怎样会是 identical 的。

Sam's answer

A sperm has 46 chromosomes and an egg has 46, so the baby gets all of them. The mother's egg decides if the baby is a boy or a girl. Identical twins form when two eggs are fertilised by two sperm.

Sam 的答案有 3 处错。找出来，改正，并说出为什么丢分。

1. a sperm and an egg have 23 chromosomes each, so the zygote has 46 (L1)
2. the sperm decides the sex: it carries X or Y, and every egg carries X (L7)
3. identical twins come from one fertilised egg that splits; two eggs give non-identical twins (L11)

1:03–1:04。三句话每句一个错，让她逐句改成正确的英文句子。

Practice

Q1 State where fertilisation happens and name the cell that is formed. [A]

Q2 A boy and a girl are twins. Are they identical twins? Explain. [M]

Q3 Explain why a child has features of both parents but is not identical to either of them. [E]

Practice 练习

Q1 说出 fertilisation 在哪里发生，以及形成的细胞叫什么。 [A]

in the oviduct; a zygote

Q2 一对双胞胎是一男一女。他们是 identical twins 吗？解释。 [M]

no: identical twins come from one zygote, so they have the same chromosomes and are the same sex; a boy and a girl must come from two eggs and two sperm

Q3 解释为什么孩子有父母双方的特征，但和任何一方都不完全一样。 [E]

23 chromosomes come from the mother's egg and 23 from the father's sperm, so the child's genes are a mixture of both; the mixture is a new combination, different from each parent

1:04–1:06. Q3 要出现数字 23 和 “mixture / new combination” 才算 E。

Classwork

In-class work · Part C

- 1 Put the stages from sex cell to foetus in order.
- 2 Work out the sex of a baby from its chromosomes.
- 3 Explain how two kinds of twins form.

Give the number of chromosomes each time you name a cell.

Classwork 课堂练习

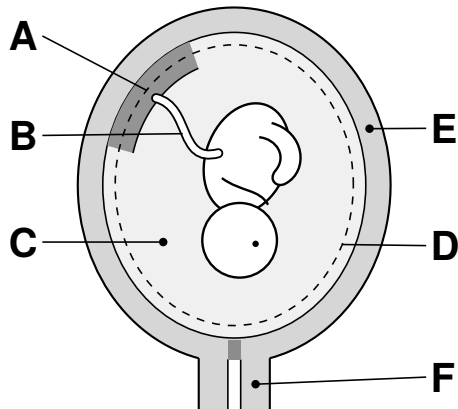
In-class work · Part C (约 6 分钟)

- 1 把从性细胞到 foetus 的各阶段排序。
- 2 根据染色体判断婴儿的性别。
- 3 解释两种双胞胎是怎样形成的。

每次提到一种细胞，都写出它的染色体数目。

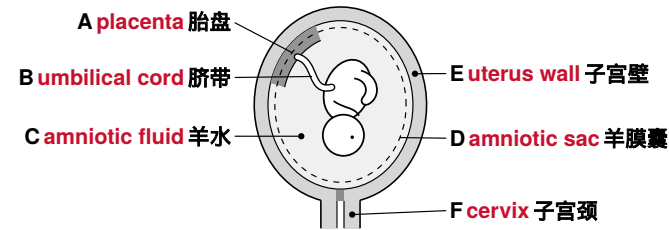
答案在 [inclass_answers.pdf](#) 的 **Part C**

Quick review: The foetus in the uterus



- 1 Name parts A to F.
- 2 Which part cushions the foetus?
- 3 Which part joins the foetus to the placenta?

Quick review 快速回忆: 子宫里的 foetus

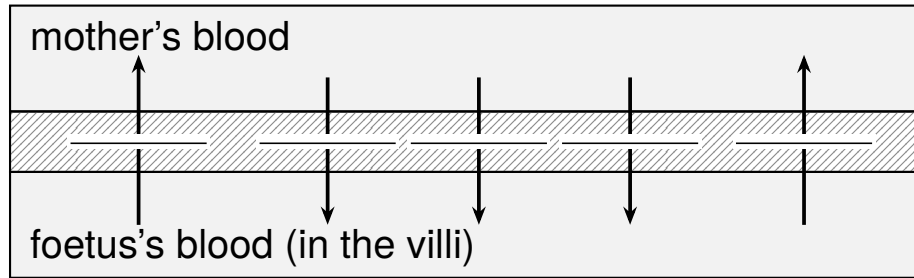


- 1 说出 A-F 的名称。
见图上红字
- 2 哪一部分给胎儿缓冲？
the amniotic fluid
- 3 哪一部分把胎儿和 placenta 连起来？
the umbilical cord

1:12-1:14。她 booklet p.25 的连线和功能表都做对了，这一页快速过。

Say it 读音
umbilical
um-BIL-ih-kul
amniotic
am-nee-OT-ik

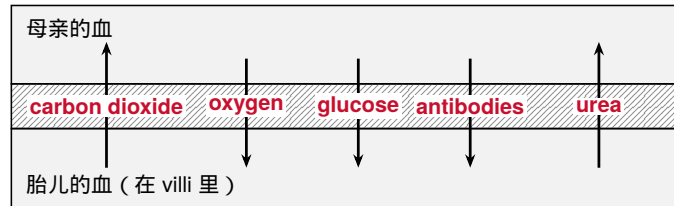
The placenta: what crosses?



Down: three things the fetus needs. Up: two waste products.

The two bloods never _____. Substances cross by _____.

Placenta 胎盘：什么能通过？



向下的箭头：胎儿需要的三样东西。向上的箭头：两种废物。

两边的血永远不 **mix**。物质靠 **diffusion**（扩散）通过。

glucose 也可以答 food / nutrients; **antibodies** 是抗体; **urea** 是尿素

1:14–1:16。易错（L9）：母亲的血不会流进胎儿，中间隔着一层薄壁。答题要写“什么物质 + 往哪个方向”。

Say it 读音

villi

VIL-eye

glucose

GLOO-kohz

antibodies

AN-tee-bod-eez

urea

yoo-REE-uh

diffusion

dih-FYOO-zhun

Workbook: the role of the placenta

The placenta develops from foetal tissues and allows the _____ of materials between mother and foetus. It has a large number of blood vessels that do not touch. This ensures there is no _____ of the maternal and foetal blood.

- a Give three reasons why it is important that the maternal and foetal blood do not mix.
- b Explain why the embryo needs oxygen and glucose.
- c Which substances diffuse from the foetus' blood into the mother's blood and why is it important that they are removed?

Workbook 练习册: placenta 的作用

Placenta 由胎儿的组织发育而来, 让母亲和胎儿之间进行物质的 **exchange (diffusion)**。它有大量互不接触的血管, 保证母亲的血和胎儿的血不会 **mixing**。

- a 说出母血和胎血不能混合的三个原因。
they can be different blood types; the mother's blood pressure is higher and would damage the foetus's small blood vessels; it stops many bacteria and harmful substances reaching the foetus
- b 解释为什么 embryo 需要 oxygen 和 glucose。
for respiration, which releases the energy its cells need to divide and grow
- c 哪些物质从胎儿的血扩散到母亲的血? 为什么必须把它们除去?
carbon dioxide and urea; they are waste products and would poison the foetus if they built up

1:16–1:19。booklet p.26 她整页空着, 是最该补的一页, 屏幕上英文原文, 让她把答案写到 booklet 上。maternal 是“母亲的”, foetal 是“胎儿的”。

Say it 读音

foetal

FEE-tul

maternal

muh-TUR-nul

respiration

res-pih-RAY-shun

Oxygen to the foetus: write it in this order

- 1 The mother's blood has a _____ concentration of oxygen than the foetus's blood.
- 2 Oxygen _____ from high to low concentration, across the thin wall.
- 3 The villi give a large _____, so diffusion is fast.
- 4 The foetus's blood carries the oxygen away in the umbilical _____.

A: line 2 **M:** lines 1–2 **E:** lines 3–4 as well

Now tell the same story for carbon dioxide.

Oxygen 怎样到胎儿：按这个顺序写

- 1 母亲的血里 oxygen 的 concentration (浓度) 比胎儿的血 higher。
- 2 oxygen 从高浓度向低浓度 diffuses, 穿过薄壁。
- 3 villi (绒毛) 提供很大的 surface area, 所以扩散得快。
- 4 胎儿的血经 umbilical vein 把 oxygen 带走。

A: 第 2 行 **M:** 第 1–2 行 **E:** 再加上第 3–4 行

现在用同样的方式讲 carbon dioxide。

the foetus's blood, arriving in the umbilical artery, has a higher concentration of carbon dioxide than the mother's blood, so carbon dioxide diffuses across the villi into the mother's blood, which carries it away

1:19–1:21。她 booklet p.24 第 4 题写得很好, 已经到 E。这里让她不看书把 oxygen 口头讲一遍, 再自己讲 carbon dioxide (L10)。

Say it 读音

concentration

kon-sun-TRAY-shun

artery

AR-tuh-ree

Harm: what else crosses the placenta?

From the mother	Effect on the foetus
cigarette smoke	less _____ reaches it; the baby is smaller
alcohol	damages the developing _____
drugs	the baby can be born addicted
rubella virus	deafness; heart and eye defects

Why can even a small amount do a lot of damage?

危害：还有什么能通过 placenta ?

来自母亲	对胎儿的影响
cigarette smoke 香烟烟雾	到达胎儿的 <u>oxygen</u> 变少；婴儿偏小
alcohol 酒精	损伤发育中的 <u>brain</u>
drugs 毒品	婴儿出生时可能已经 addicted (成瘾)
rubella virus 风疹病毒	耳聋；心脏和眼睛的缺陷

为什么很少的量也会造成很大的伤害？
the foetus is very small and is growing very quickly

1:21–1:23。香烟：carbon monoxide 占了母亲血里 oxygen 的位置，nicotine 也能通过 placenta。booklet p.29 的 Research Task 是做海报，考试记住这四行就够。

Say it 读音
rubella
roo-BEL-uh
addicted
uh-DIK-tid
monoxide
mon-OK-syd
nicotine
NIK-uh-teen

Pregnancy in numbers

- 1 Pregnancy (gestation) lasts about _____ weeks, counted from the first day of the last period.
- 2 The embryo is called a foetus from about week _____.
- 3 Losing a foetus before _____ weeks is a miscarriage.
- 4 A baby born before _____ weeks is premature.

A baby is born at 35 weeks. Which word describes this birth?

怀孕里的几个数字

- 1 怀孕 (gestation) 大约 **40** 周，从最后一次月经的第一天算起。
- 2 大约从第 **8** 周起，embryo 改叫 foetus。
- 3 **20** 周之前失去胎儿，叫 miscarriage (流产)。
- 4 **38** 周之前出生，叫 premature (早产)。

一个婴儿在 35 周出生。用哪个词描述？

premature

1:23–1:24。第 2 行：glossary 写 after 8 weeks，p.28 写 nine weeks，答 8 或 9 都对。20 周以后失去胎儿叫 stillbirth。

Say it 读音

gestation

jes-TAY-shun

miscarriage

MIS-ka-rij

premature

PREM-uh-chur

Birth: three stages

Put the cards in order. Then name each stage.

- A** The placenta is pushed out.
- B** Muscles of the uterus contract. The cervix opens (dilates). The waters break.
- C** The baby is pushed out through the vagina. The cord is tied and cut.

Stage 1 _____ Stage 2 _____ Stage 3 _____

What are “the waters”?

Birth 出生：三个阶段

把卡片排好顺序，再说每个阶段的名称。

- A** placenta 被排出。
- B** uterus 的肌肉收缩；cervix 张开 (dilates)；羊水破了。
- C** 婴儿经 vagina 被推出；脐带结扎后剪断。

顺序：**B** → **C** → **A**

Stage 1 labour Stage 2 delivery Stage 3 afterbirth

“the waters” 是什么？
the amniotic fluid

1:24–1:26。labour 是阵痛和宫颈张开，delivery 是婴儿娩出，afterbirth 是排出胎盘。

Say it 读音
dilates
dy-LAYTS
labour
LAY-bur

Workbook: fertility problems and solutions

Problems	Solutions
Low sperm count	Injecting sperm cell into egg cell, then IVF
Oviducts too narrow	
Sperm ducts blocked	
Lining stops implantation	
Few ripe eggs released	
Sperm do not swim well	

Workbook 练习册：不孕的问题和解决办法

Problems 问题	Solutions 解决办法
Low sperm count 精子数量少	Injecting sperm cell into egg cell, then IVF (书上已给)
Oviducts 太窄或受损	doctors try to widen the oviducts; or IVF
Sperm ducts 堵塞	doctors try to widen the sperm ducts
内膜阻止 implantation	IVF
排出的成熟卵太少	IVF
Sperm 游不好	inject a sperm cell into an egg cell with a fine needle

1:26–1:28. booklet p.32 的空表，答案都在 p.31 的两段文字里。让她一边在 p.31 划线，一边填表。第 3 行答“IVF”是书上的原话。

IVF: fertilisation in a dish

Put the cards in order.

- A** Eggs are collected from the woman's ovaries.
- B** An embryo is put into the uterus, where it may implant.
- C** The woman is given hormones, so several eggs mature.
- D** The eggs are mixed with sperm in a dish. Some are fertilised.
- E** The embryos grow for about two days.

Why are twins more common after IVF?

IVF: 在培养皿里受精

把卡片排好顺序。

- A** 从女性的 ovaries 里取出 eggs。
- B** 把一个 embryo 放进 uterus，它可能在那里 implant。
- C** 给女性用 hormones，让多个 eggs 成熟。
- D** eggs 和 sperm 在培养皿里混合，其中一些受精。
- E** embryos 培养大约两天。

顺序：C → A → D → E → B

为什么 IVF 之后双胞胎更常见？

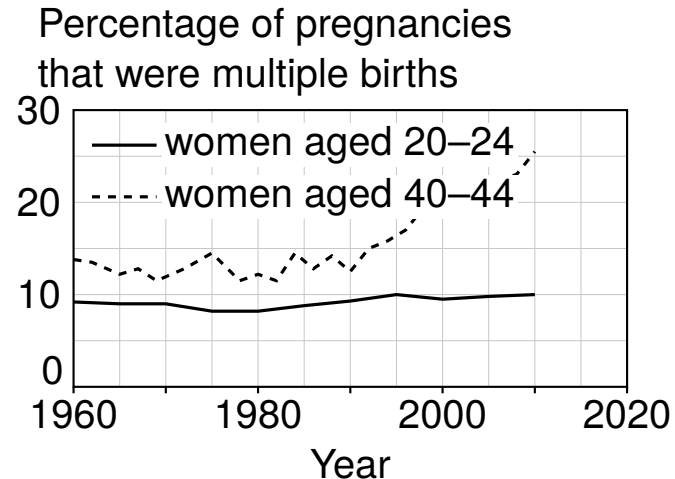
often more than one embryo is put into the uterus, and each one can implant

1:28–1:30。in vitro 的意思是“在玻璃里”，指受精发生在体外。易错（L14）：婴儿不是在试管里长大的，embryo 要放回 uterus。

Say it 读音

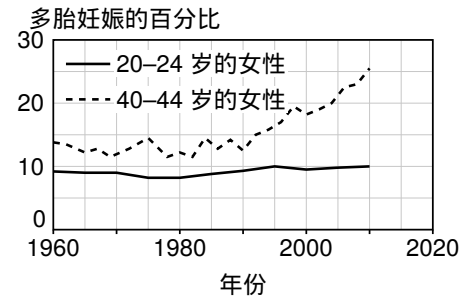
vitro
VEE-troh

Workbook: question 3, describe



- (a) Describe what has happened to the number of multiple births in women aged 40–44.
- (c) Describe what happened to the number of multiple births in women aged 20–24.

Workbook 练习册：第 3 题，describe



- (a) 描述 40–44 岁女性多胎妊娠数量的变化。
from 1960 to about 1990 it stayed between about 11 % and 15 %; after 1990 it rose steadily to about 25 % in 2010
- (c) 描述 20–24 岁女性多胎妊娠数量的变化。
it stayed almost the same, at about 9 % to 10 %, from 1960 to 2010

1:30–1:32. Describe 要写趋势，加上从图上读出的数字和年份，不要写原因 (L12)。虚线是 40–44 岁，实线是 20–24 岁。

Workbook: suggest and predict

Headlines: “Fertility treatment shown to increase chances of multiple birth” · “Older women more likely to have problems becoming pregnant” · “Fertility treatment becoming more common”

Q3 (b) Suggest a hypothesis for why this has been happening.

Q3 (d) Suggest a hypothesis for why this pattern is not like that for women aged 40–44.

Q4 (a) Predict the percentage of multiple births last year for women aged 40–44.

Workbook 练习册: suggest 和 predict

新闻标题: “生育治疗会增加多胎的机会” · “年龄大的女性更容易有怀孕困难” · “生育治疗越来越普遍”

Q3 (b) 提出一个假设 (hypothesis), 解释为什么会出现这种变化。

fertility treatment has become more common; older women are more likely to need it; fertility treatment increases the chance of a multiple birth

Q3 (d) 提出一个假设, 解释为什么 20–24 岁的规律和 40–44 岁不一样。

women aged 20 to 24 rarely have problems becoming pregnant, so few of them have fertility treatment

Q4 (a) 预测去年 40–44 岁女性多胎妊娠的百分比。

about 30 % or a little more: any value above 25 % that continues the rising line. (b) I extended the trend line from 1990 to 2010 forward to last year

1:32–1:35. Suggest 要用上三条新闻标题里的信息。booklet p.35 第 1 题: a multiple birth is two or more babies from one pregnancy. 第 2 题: the hormones make the ovaries release more eggs, so there is more chance that one is fertilised.

Say it 读音

hypothesis

hy-POTH-uh-sis

Spot the error (4)

Describe how a foetus gets oxygen and how it is protected in the uterus.

Sam's answer

The mother's blood flows through the umbilical cord into the foetus. Oxygen moves by diffusion from a low concentration to a high concentration. The amniotic fluid gives the foetus its food and protects it from bumps.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Spot the error 找错 (4)

题目：描述胎儿怎样得到 oxygen，以及它在 uterus 里怎样受到保护。

Sam's answer

The mother's blood flows through the umbilical cord into the foetus. Oxygen moves by diffusion from a low concentration to a high concentration. The amniotic fluid gives the foetus its food and protects it from bumps.

Sam 的答案有 3 处错。 找出来，改正，并说出为什么丢分。

1. the two bloods never mix: only the foetus's blood flows in the umbilical cord (L9)
2. diffusion is from a high concentration to a low concentration (L10)
3. the amniotic fluid does not feed the foetus: food comes from the mother's blood through the placenta (L13)

1:35–1:36. “protects it from bumps” 是对的。

Practice

- Q1** Name two substances that pass from the foetus's blood to the mother's blood. [A]
- Q2** Explain why a pregnant woman who smokes is likely to have a smaller baby. [M]
- Q3** IVF clinics now usually put one embryo into the uterus, not three. Suggest why, and explain how this changes the number of twins and triplets. [E]

Practice 练习

- Q1** 说出两种从胎儿的血进入母亲的血的物质。 [A]
carbon dioxide and urea
- Q2** 解释为什么吸烟的孕妇更可能生下偏小的婴儿。 [M]
carbon monoxide from the smoke is carried in the mother's blood, so less oxygen crosses the placenta; the foetus's cells respire less and release less energy, so it grows more slowly
- Q3** 现在做 IVF 通常只放一个 embryo 进 uterus, 而不是三个。说出可能的原因, 并解释这怎样改变双胞胎和三胞胎的数量。 [E]
a multiple birth is riskier for the mother and the babies, who are often premature and small; with one embryo only one can implant, so there are fewer twins and triplets

时间紧就只做 Q1 和 Q2, Q3 口头说。Q2 的链条: less oxygen, less respiration, less energy, slower growth。

Classwork

In-class work · Part D

- 1 Label the foetus in the uterus.
- 2 Explain how a waste product leaves the foetus.
- 3 Describe and explain a table about the foetus.

Name the substance, the direction and the process.

Classwork 课堂练习

In-class work · Part D (约 6 分钟)

- 1 给子宫里的 foetus 图标注。
- 2 解释一种废物怎样离开胎儿。
- 3 描述并解释一张关于 foetus 的数据表。

要写出：什么物质、往哪个方向、什么过程。

答案在 **inclass_answers.pdf** 的 **Part D**

Ways to lose marks

Say the correct version of each one.

- 1 “A sperm cell has 46 chromosomes.”
- 2 “Fertilisation happens in the uterus.”
- 3 “Ovulation is when the lining is lost.”
- 4 “The mother’s blood flows into the foetus.”
- 5 “A sperm has a tail.” (asked to explain an adaptation)
- 6 “It went up because of IVF.” (asked to describe a graph)

Ways to lose marks 失分点

说出每一句的正确版本。

- 1 “A sperm cell has 46 chromosomes.”
23: every sex cell has one set; the zygote and body cells have 46 (L1)
- 2 “Fertilisation happens in the uterus.”
in the oviduct; the embryo implants in the uterus later (L2)
- 3 “Ovulation is when the lining is lost.”
ovulation is the release of an egg; losing the lining is menstruation (L5)
- 4 “The mother’s blood flows into the foetus.”
the bloods never mix; oxygen and food diffuse across the placenta (L9)
- 5 “A sperm has a tail.” (题目要求解释 adaptation)
add how it helps: so that it can swim to the egg (L3)
- 6 “It went up because of IVF.” (题目要求 describe 图表)
describe needs the trend and numbers; the reason is for suggest or explain (L12)

Dr. Ryan Lei · Auckland 50

1:42–1:43. 每句让她直接说出正确版本，不展开。

Summary

- 1 Sex cells have 23 chromosomes; fertilisation in the oviduct makes 46.
- 2 Cycle: period, lining builds up, ovulation on day 14, lining kept thick.
- 3 Placenta: diffusion from high to low; the two bloods never mix.
- 4 Explain = fact + because. Describe a graph = trend + numbers.

Summary 总结

- 1 Sex cells 有 23 条染色体；在 oviduct 里 fertilisation 之后是 46 条。
- 2 周期：period，内膜 builds up，第 14 天 ovulation，内膜保持 thick。
- 3 Placenta：diffusion，从高浓度到低浓度；两边的血永远不混合。
- 4 Explain = fact + because。Describe a graph = trend + numbers。

Mini mock

In-class work · Part E

12 minutes of writing, no notes, no booklet.

Then we mark it together: A, M or E for each question.

Read the command word first: state, describe, explain, suggest.

Mini mock 小模考

In-class work · Part E

写 12 分钟，不看笔记，不看 booklet。

然后一起批改：每道题给 A、M 或 E。

先看指令词：state、describe、explain、suggest。

答案在 **inclass_answers.pdf** 的 **Part E**

1:44–1:56 写，1:56–1:59 对答案。state 只写事实；describe 写现象或趋势；explain 要有 because；suggest 是有依据的推测。

Homework

Homework: two parts, about 40 minutes each

- 1 Part 1: sex cells, the two systems, the menstrual cycle.
- 2 Part 2: fertilisation, chromosomes, pregnancy, fertility.

Do the two parts on two different days.

Homework 作业

Homework: 两部分，每部分约 40 分钟

- 1 Part 1: 性细胞、男女生殖系统、月经周期。
- 2 Part 2: 受精、染色体、怀孕、不孕。

两部分分两天做。

答案在 [homework_answers.pdf](#)

1:59–2:00。发 Homework。In-class work 收回，下次课前看她哪几个 Part 的 E 没拿到。